

AZKA AFTAB

☎ +60176491858 ✉ azka.aftab25@gmail.com 🌐 <https://www.linkedin.com/in/azka-aftab-59b027376/>

🌐 <https://azka-aftab-25.vercel.app/> 🌐 <https://github.com/Azzkaa> 📍 Location: Saudi Arabia, Tabuk

Skills

Languages: Python, C++, Java, TypeScript, SQL, PHP

Frameworks & Libraries: TensorFlow, PyTorch, Django, Spring Boot, Laravel, Vue.js, Node.js

Tools : AWS, Git, Firebase, Power BI, Tableau, WEKA, Orange3

Work Experience

Student Research Assistant

02/2025 - 07/2025

Universiti Teknologi Malaysia

Damai, Kuala Lumpur

- Preprocessed thermal image datasets and researched a **deep learning-based thermal fault detection model for TNB** using Python (MobileNetV2, TensorFlow, PyTorch) for fault detection and SAM for segmentation.
- Conducted data cleaning, model evaluation, and visualization, ensuring accurate interpretation of results.
- Achieved **93 % classification accuracy** and **0.92 IoU** on real-world substation datasets (FLIR, Fluke).
- Developed a modular AI pipeline integrating classification, segmentation, and ΔT scoring.
- Collaborated with supervisors on technical writing and system performance analysis.

Technical Project Management Internship

08/2024 - 02/2025

Fireworks Solutions Sdn Bhd

Petaling Jaya, Selangor

- Optimized database schemas and backend data pipelines to ensure high data integrity for production systems.
- Managed version control and technical documentation for data workflows across 5 concurrent projects.
- Analyzed bugs through backend data and logs, used insights to build enhanced UI and logic updates.
- Translated client requirements (SOGO, Focus Point) into technical data specifications and performance KPIs.

Projects

- [AI-Powered Gmail Auto-Responder with Telegram Approval \(Automation/ETL Workflow\)](#) – Designed an n8n-based workflow integrating Gmail API, AI model, and Telegram Bot for automated email classification, approval, and response. Demonstrated skills in ETL pipeline design, API integration, and workflow automation.
- [Handwritten Digit Recognition \(AI/ML\)](#) – Low-latency digit classification with preprocessing and model optimization.
- [Thermal Fault Detection \(Research Project\)](#) – Deep learning models for substation anomaly detection.
- [Penguin Species Clustering and Analysis System](#) – Developed an unsupervised learning system using K-Means, DBSCAN, and Agglomerative Clustering with PCA dimensionality reduction to classify penguin species.
- [SP Course Assignment Tracking System \(Web\)](#) – Java + SQL backend with authentication, CRUD operations, and project tracking.
- [Application Development SKSU Student Portal \(Web\)](#) – PHP/MySQL-based system for student management.

Education

Universiti Teknologi Malaysia

CGPA: 3.89/4.00

Bachelor of Computer Science (Software Engineering) with Honors

November 2025

Certifications/Extra Curriculars

BRUNCloud Young Technovation (RCYT) Techlympics

AWS Cloud Practitioner

Achievement: IBIM-INCEIF GIFF22 HACKATHON - 2nd Runner UP

Vice President, Sakura Kai Club | 1 Year